CCC-C-443a
OCTOBER 20, 1960
SUPERSEDING
Int. Fed. Spec. CCC–C–00443 (GSA–FSS)
June 22, 1960 and
Fed. Spec. CCC–D–761
February 7, 1933
(See 6.4)

FEDERAL SPECIFICATION

CLOTH, COTTON, DUCK; (SINGLE AND PLIED FILLING YARNS, FLAT)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification covers the requirements of unbleached flat duck utilized in the fabrication of tentage and equipage items.

1.2 Classification.

1.2.1 *Types.*—The cloth, cotton, flat duck shall be of the following types as specified (see 6.2):

Type I.—Single filling.
Type II.—Double filling (plied).

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards.—The following specifications and standards of the issues in effect on date of invitation for bids form a part of this specification:

Federal Specifications:

CCC–T–191—Textile Test Methods.
PPP–P–51 — Packaging, Packing and Marking for Textile Fabrics (Woolens, Worsteds, Cottons, Silks and Synthetics).

Federal Standards:

Fed. Std. No. 4—Glossary of Fabric Imperfections.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Specifications:

MIL–B–3149 — Baling of Clothing and Equipage.

Military Standards:

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. — The following publication of the issue in effect on date of invitation for bids forms a part of this specification:

Textile Fiber Products Identification Act.

(Copies may be obtained free of charge from the Federal Trade Commission, Washington 25, D. C.)

3. REQUIREMENTS

3.1 Material.

3.1.1 *Yarns.* — Yarns shall be made from cotton which has been cleaned, carded, drawn and spun into single yarns for type I and twisted into plied yarns for type II.

TABLE I.—*Physical requirements, type I, single filling*

Weight per square yard (minimum)	Yarns per inch (minimum)		Breaking strength[2] pounds (minimum)		Air permeability (cubic ft/min/ft²) average (maximum)	Weave
Ounces	Warp[1]	Filling	Warp	Filling		
7.26	74	27	90	55	--	
8.47	74	27	100	65	--	
9.68	74	27	130	85	--	Oxford,
10.89	74	26	135	95	--	2 ends
12.10	72	26	160	105	--	weaving
13.32	72	26	165	110	--	as one
14.53	72	25	170	120	--	(plain
14.90	76	28	180	160	--	weave)
17.55	86	27	270	245	4	
18.15	72	25	180	140	3	

[1] Two yarns woven as one.

[2] A minus tolerance of 5 percent in breaking strength will be permitted in any one direction provided the combined strength equals the sum of the specified warp and filling requirements.

TABLE II.—*Physical requirements, type II, double filling (plied)*

Weight per square yard (minimum)	Yarns per inch (minimum)		Ply	Breaking strength[2] pounds (minimum)		Weave
Ounces	Warp[1]	Filling	Filling	Warp	Filling	
7.26	86	30	2	100	60	
8.47	86	30	2	115	85	Oxford,
9.68	86	30	2	125	100	2 ends
10.89	84	30	3	140	110	weaving
12.10	84	30	3	150	115	as one
13.32	84	29	4	165	135	(plain
14.53	82	29	4	180	160	weave)
18.15	82	25	5	195	190	

[1] Two yarns woven as one.

[2] A minus tolerance of 5 percent in breaking strength will be permitted in any one direction provided the combined strength equals the sum of the specified warp and filling requirements.

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3.2 Physical requirements.—The cloth, cotton duck covered by this specification shall conform to the requirements specified in tables I and II when tested as specified in 4.2.2.

3.3 Width.—The width inclusive of selvage shall be as specified (see 6.2). A tolerance of minus ¼ inch will be allowed on the specified width.

3.4 Nonfibrous material content. — The cloth, cotton duck shall contain not more than 12 percent total size, finishing or other nonfibrous material when tested as specified in 4.2.2.

3.5 Fiber identification.—Each piece or roll shall be labeled, ticketed, or invoiced for fiber content in accordance with the Rules and Regulations of the Textile Fiber Products Identification Act.

3.6 Put-up.

3.6.1 *Folded pieces.* — When folded pieces are specified (see 6.2) pieces 80 yards or more in length shall be delivered. Pieces less than 80 yards will be acceptable up to 10 percent of the contract quantity, provided no piece is less than 20 yards in length, and not more than two pieces less than 80 yards shall be included in any one package except where otherwise specified (see 6.2).

3.6.2 *Rolls.*—When roll packing is specified (see 6.2), pieces 80 yards or more in length shall be delivered. Pieces less than 80 yards will be acceptable up to 10 percent of contract quantity, provided no piece is less than 20 yards in length and not more than two pieces less than 80 yards are included in any one roll, except where otherwise specified (see 6.2). The pieces shall be sewn together with a butted-type seam on all duck 14.90 ounces per square yard and lighter in weight, using cotton thread and shall have a minimum of 9 stitches per inch. The stitching shall be placed not closer than ½ inch from the cut or raw edge of the cotton duck. On duck heavier in weight than 14.90 ounces per square yard, the pieces will be lapped and not sewn together. The number of pieces in the roll will be indicated by stencil marking on the last 6 inches of the roll on the outer surface (see 6.2).

3.7 Workmanship.—The cloth, cotton duck shall conform to the quality and grade of product established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Contractor's inspection responsibility. —Unless otherwise specified, the supplier is responsible for the performance of all inspection requirements prior to submission for Government inspection and acceptance. Except as otherwise specified, the supplier may utilize his own facilities or any other commercial laboratory acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order.

4.2 Inspection for acceptance. — Except when otherwise specified by Civil Government agencies, inspection shall be in accordance with the provisions of Military Standard MIL–STD–105, except where otherwise indicated.

4.2.1 *Examination of the end item.* — The cloth shall be examined to determine conformance with the requirements of this specification. Defects found during this examination shall be classified in accordance with 4.2.1.1, 4.2.1.2 and 4.2.1.3.

4.2.1.1 *Yard by yard examination.* — All defects listed in table III shall be counted regardless of their proximity of one to another except where two or more defects represent a single local condition of the cloth in which case only the more serious defect shall be

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TABLE III.—*Defects*

Yard-by-yard examination

Glossary numbers of Fed. Std. No. 4	Defects	Classification	
		Major	Minor
55, 56, 90, 91, 92	Hole, cut, tear or break (including selvage)—any	X	
78	Smash—any	X	
79, 89	Abrasion mark, bruise, tender or weak spot—any	X	
11, 33	Broken or missing yarns:		
	Three or more contiguous regardless of length	X	
	Two contiguous, 2 inches or more missing	X	
	Two contiguous, less than 2 inches or single 2 inches or more missing		X
9	Fine or light filling bar, light place clearly visible*	X	
2	Coarse filling bar or heavy place clearly visible* and exceeding 2 linear inches		X
5, 27, 29, 77	Slough off, jerked in filling, slubs, gouts, knots—more than three times the thickness of the normal yarns		X
9, 12	Open place or crack clearly visible*	X	
13, 34	Floats:		
	—Multiple ½ inch or more in combined warp and filling directions, or harness skip	X	
	—Multiple less than ½ inch in combined warp and filling directions, or single 2 inches or more in length		X
41, 42	Reed mark or temple mark resulting in a clear warpwise separation markedly wider than normal		X
40	Wrong draw—½ yard or more		X
49, 50	Slack or tight selvage resulting in pronounced slackiness or waviness in cloth		X
51	Scalloped selvage—indentations exceeding ½ inch		X
48	Curled or folded selvage—½ yard or more		X
53, 54	Stringy, beaded, corded, loopy or rough selvages—½ yard or more, resulting in raised places along the selvage edge 2 or more times the normal fabric thickness, ½ yard or more		X
74, 75, 76	Spot, stain or streak—covering more than 1 inch in combined directions clearly visible*	X	
	¼ inch but nor more than 1 inch in combined directions, clearly visible*		X

* At normal inspection distance (approximately 3 feet).

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counted. A continuous defect shall be counted as one defect for each warpwise yard or fraction thereof in which it occurs. The unit of product for this examination shall be 1 linear yard. The acceptable quality level for this examination shall be as follows:

1. Width of fabric 56 inches or less: 0.40 major defects and 2.5 total defects (major and minor combined) per 100 units (yards).
2. Width of fabric more than 56 inches: 0.65 major defects and 4.0 total defects (major and minor combined) per 100 units (yards).

Unless otherwise specified, the inspection level shall be level II of Military Standard MIL–STD–105. The lot size shall be expressed in terms of 1 yard each. An approximately equal number of yards shall be examined from each bolt or roll in the sample. The number of rolls or folded pieces from which the sample is to be selected shall be in accordance with table IV.

4.2.1.2 *Overall examination.*—Each defect listed below shall be counted not less than once in each piece examined. The unit of product for the examination shall be one piece. The sample size and acceptance number shall be as shown in table IV.

Defects

Poorly constructed—not firmly or tightly woven, overall uncleanliness

Baggy, ridgy or wavy cloth—clearly visible*

Trash, motes, seed coat, fragments, etc. —clearly visible*

Rancid or otherwise objectionable odor

Width not within specified tolerances

* At normal inspection distance (approximately 3 feet).

TABLE IV.—*Sample size*

Lot size in yards	Sample size in pieces	Maximum number of defects acceptable in sample
Up to 1,300[1]	3	0
1,301 to 3,200	5	0
3,201 to 8,000	7	0
8,001 to 22,000	10	0
22,001 to 110,000	15	1
110,001 and over	25	1

[1] If a lot contains fewer than 3 pieces, each piece in the lot shall be examined.

4.2.1.3 *Length examination.*

4.2.1.3.1 *Examination for length of individual pieces.*—Each piece in the sample shall be examined for gross length. Any gross length found to be less than the specified minimum length or more than maximum gross length permitted, or any gross length found to be more than 2 yards less than the gross length marked on the ticket shall be considered a defect with respect to length. The unit of product for this examination shall be one piece. The sample size and acceptance number shall be as shown in table IV.

4.2.1.3.2 *Examination for total yardage in sample.*—The lot shall be unacceptable if the total of the actual gross lengths of the pieces in the sample is less than the total gross lengths marked on the piece tickets. The pieces examined shall be those selected for examination of individual pieces.

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4.2.2 *Testing of the end item.*—The physical and chemical values specified in Section 3 apply to the average of the determinations made on a unit of product for test purposes as specified in the applicable test methods. The methods of testing specified in Federal Specification CCC–T–191 wherever applicable and as listed in table V shall be followed. The unit of product for testing shall be 1½ yards full width of the cloth. For each characteristic, the acceptable quality level shall be 6.5 test failures per 100 units, and the sample size shall be in accordance with inspection level L–1 of Military Standard MIL–STD–105. The lot size shall be expressed in terms of 1 yard each.

4.3 Examination of preparation for delivery requirements.—An examination will be made to determine that packaging, packing and marking requirements of Section 5 of this specification are complied with. Defects will be scored in accordance with the list below. The sample unit shall be one roll, bale or container (as applicable) fully prepared for delivery. The lot size shall be the number of completely packed rolls, bales or containers (as applicable) in the inspection lot. The inspection level shall be level L–4 and the AQL shall be 4.0 defects per hundred units.

Examine	Defect
Marking (applicable to all levels of packing)	Omitted, incorrect, illegible; of improper size, location, sequence or method of application.
Packaging (applicable to all levels)	Not packaged as specified.
Packing (applicable to rolls only)	Closure of cloth covered bundles incorrect. Any component missing or damaged, affecting serviceability. Insecurely sewn bundles, affecting serviceability.
Packing (applicable to bales only)	To be classified in accordance with Military Specification MIL–B–3149.
Packing (applicable to containers only)	Inadequate application of components such as: incomplete closure of case liners, container flaps, loose strapping, inadequate stapling. Bulging or distortion of containers.
Weight (exterior and interior)	Gross/net weight exceeds requirements.

5. PREPARATION FOR DELIVERY

5.1 Packaging.—Packaging shall be level A, B or C as specified (see 6.2 and 6.2.1).

5.1.1 *Levels A, B, and C.*—The cotton duck cloth of one type only shall be packaged in accordance with the applicable requirements for class 4 fabrics of Federal Specification PPP–P–51.

5.2 Packing.—Packing shall be level A, B or C as specified (see 6.2 and 6.2.1).

5.2.1 *Levels A, B and C.*—The cotton duck cloth of one type only shall be packed in accordance with the applicable requirement for class 4 fabrics of Federal Specification PPP–P–51.

5.3 Marking. — In addition to any special marking required by the contract or order, shipments shall be marked in accordance with the requirements of Federal Specification PPP–P–51.

6. NOTES

6.1 Intended use.—The cloth, cotton duck, single and plied filling yarns (flat) covered by this specification are utilized in the fabrication of tentage and related items. The type I, 14.90 and 17.55 ounce per square yard flat ducks are specifically designed as emergency substitutes for type III, 12.29 ounce Army duck and type I, No. 8 duck respectively, covered by Federal Specification CCC–C–419.

6.2 Ordering data. — Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

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(*a*) Title, number and date of this specification.
(*b*) Type and weight of duck required (see 1.2. and 3.2).
(*c*) Width of duck required (see 3.3).
(*d*) Put-up, either folded pieces or rolls (see 3.6).
(*e*) Put-up, other than specified (see 3.6).
(*f*) Whether the cloth is to be packaged, packed and marked level A, level B or level C (see 5.1, 5.2 and 5.3).

6.2.1 In the preparation of contracts or orders it must be noted that various options, choices and alternatives as indicated in Federal Specification PPP–P–51 may be exercised in the preparation for delivery of cloth.

6.3 Short pieces.—Pieces less than 80 yards in length in excess of the amount specified in 3.6 and pieces shorter in length than 20 yards may be presented to the contracting officer for consideration for end users where short length pieces may be considered.

TABLE V.—*Test methods*

Characteristic	Req. ref.	Test method	Number of determinations per individual unit of product	Results reported as:
Weight	Tables I and II	5041	5	Avg. of 5 determinations to nearest 0.01 ounce
Weave	Tables I and II	Visual	--	Pass or fail
Breaking strength:				
Warp	Tables I and II	5100	5	Avg. of 5 determinations to nearest 1.0 lb.
Filling	Tables I and II	5100	5	Avg. of 5 determinations to nearest 1.0 lb.
Yarn ply, types I and II:				
Warp	3.1.1 and Table II	Visual	1	Pass or fail
Filling	3.1.1 and Table II	Visual	1	Pass or fail
Yarns per inch:				
Warp	Tables I and II	5050	5	Avg. of 5 determinations to nearest whole number
Filling	Tables I and II	5050	5	Avg. of 5 determinations to nearest whole number
Nonfibrous[1] materials content	3.4	2611	2	Avg. of 2 determinations to nearest 0.1 percent
Air permeability	Table I	5450	5	Avg. of 5 determinations to nearest 0.1 cubic feet/minute/square foot of cloth
Identification of cotton	3.1.1	1200	1	Pass or fail

[1] The amount of nonfibrous materials shall be determined with the following formula:

$$\text{Total nonfibrous materials, percent} = \frac{O-S}{O} \times 100$$

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6.4 Supersession data.—This specification includes the requirements of Military Specification MIL–D–3396 Duck, Cotton, Single and Plied Filling Yarns (Flat) dated 23 January 1951 and Federal Specification CCC–D–761, Duck; Cotton, Plied-Filling-Yarns and Single-Yarns (Flat Duck) dated 7 February 1933.

6.5 Color and treatment.

6.5.1 Cloth, Cotton, Duck to be fire, water, weather and mildew resistant treated should be treated under the provisions of Federal Specification CCC–C–428.

6.5.2 Cloth, Cotton, Duck to be aftertreated in any of the following conditions should be treated under the provisions of Federal Specification CCC–D–950.

(*a*) Bleached.
(*b*) Dyed.
(*c*) Water repellent aftertreated.
(*d*) Mildew resistant aftertreated.
(*e*) Water repellent and mildew resistant aftertreated.

6.6 Level B. — Packaging and packing should be used by ordering agencies only when shipment is known to be made to a holding supply depot or warehouse from which partial shipments will be made to ultimate consumers.

6.7 Transportation description. — The transportation descriptions and minimum weights applicable to this item are:

Rail:
Cotton piece goods in the original piece.
Carload minimum weight, any quantity.

Motor:
Cotton piece goods not otherwise indexed.
Motor volume minimum weight, any quantity.

Notice. — When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded, by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTERESTS:
Army—C E M O Q Sig T
Navy—S CG MC Sh W.

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